

Exploratory Data Analysis of Used Cars

Cars Dataset – Data Quality, Statistical Analysis & Visual EDA

Student	Siri Havalдар
Project	Cars EDA Project
Dataset	cardekho.csv (referenced by the submitted notebook)
Date	23 August 2026
Batch	Not specified in the submitted notebook

This report is based on the analysis recorded in the submitted Jupyter Notebook. It summarizes the workflow and evidence rather than reproducing the full code.

1. Objective

The objective of this exploratory data analysis (EDA) is to understand the structure and quality of the used-car dataset, prepare the data for analysis, summarize important numerical and categorical variables, and investigate relationships between vehicle characteristics and selling price. The notebook examines distributions, category frequencies, price relationships and numerical correlations.

2. Dataset Overview

Item	Evidence from notebook
Working dataset size	6,926 rows × 12 columns
Columns	name, year, selling_price, km_driven, fuel, seller_type, transmission, owner, mileage(km/ltr/kg), engine, max_p
Numerical fields	year, selling_price, km_driven, mileage, engine, seats; max_power becomes numeric after cleaning
Categorical/text fields	name, fuel, seller_type, transmission, owner
Source reference	cardekho.csv; a public URL/source citation is not recorded in the notebook

Feature descriptions

Feature	Meaning / analytical role
name	Vehicle model/name; high-cardinality text field.
year	Vehicle year.
selling_price	Observed selling price; primary outcome examined.
km_driven	Distance driven in kilometres.
fuel	Fuel category: Diesel, Petrol, LPG or CNG.
seller_type	Seller category: Individual, Dealer or Trustmark Dealer.
transmission	Manual or Automatic.
owner	Ownership category.
mileage	Mileage in km/ltr/kg.
engine	Engine capacity.
max_power	Maximum power; originally stored as text and converted to numeric.
seats	Number of seats.

3. Data Quality and Cleaning

The notebook checks missing values, duplicate rows, categorical values and data types. It then fills missing numerical values with medians, converts max_power from object/text to numeric, removes rows outside IQR bounds for numerical columns, and finally label-encodes categorical columns. For this report, the original semantic labels are retained when describing the results.

Column / check	Recorded issue	Treatment in notebook
mileage(km/ltr/kg)	221 missing	Filled with median
engine	221 missing	Filled with median
max_power	215 missing; stored as object	Converted to numeric; resulting missing values filled with median
seats	221 missing	Filled with median
Other fields	0 missing in recorded check	No missing-value treatment needed
Duplicates	Recorded check = 0; later comment says 1,202 duplicate rows removed	drop_duplicates() executed; exact count is not reliably documented
Outliers	Extreme values visible in numerical fields	Sequential IQR filtering applied to all numerical columns

Before / after cleaning record

Stage	Rows	Columns	Basis
Working dataset before cleaning	6,926	12	df.shape output
After missing-value treatment	6,926	12	No rows removed for missing values
After sequential IQR filtering	4,836	12	df.info() output after outlier filtering

Documentation note: The notebook is internally inconsistent about duplicates. The recorded duplicate-check output is 0, while a later code comment says that 1,202 duplicate rows were removed. The displayed row index also reaches 8,125 although df.shape reports 6,926 rows. This report therefore uses the explicit dataframe outputs for the before/after cleaning record and does not invent an original-source count.

Cleaning rationale

Median imputation was used for missing numerical fields because it is less sensitive to extreme values than the mean. Converting max_power to numeric makes the field suitable for statistical analysis. IQR filtering was used to reduce the influence of extreme observations. Because the notebook applies the filter sequentially across all numerical columns, the reduction to 4,836 rows should be considered when interpreting later results.

4. Descriptive Analysis

Before outlier filtering, the notebook's numerical summary shows broad variation in price and usage. After missing-value treatment, the median selling price is ₹400,000 and the mean is approximately ₹517,271, indicating an upper tail of higher-priced vehicles. The median kilometres driven is 70,000 km and the median mileage is 19.44 km/ltr/kg.

Variable	Mean	Median	Min	Max
Year	2013.42	2014	1983	2020
Selling price	₹517,271	₹400,000	₹29,999	₹10,000,000
Km driven	73,996	70,000	1	2,360,457
Mileage	19.46	19.44	0	42
Engine	1,430.89	1,248	624	3,604
Max power	87.55	81.83	0	400
Seats	5.42	5	2	14

Categorical summary

Variable	Unique categories	Most frequent category	Frequency
Fuel	4	Diesel	3,755
Seller type	3	Individual	6,218
Transmission	2	Manual	6,342
Owner	5	First Owner	4,242
Name	2,058	Maruti Swift Dzire VDI	118

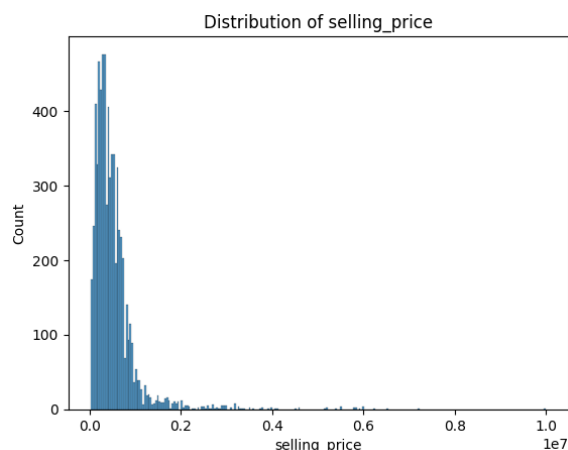
Sample records

Model	Year	Price	Km	Fuel	Transmission
Maruti Swift Dzire VDI	2014	₹450,000	145,500	Diesel	Manual
Skoda Rapid 1.5 TDI Ambition	2014	₹370,000	120,000	Diesel	Manual
Honda City 2017-2020 EXi	2006	₹158,000	140,000	Petrol	Manual
Hyundai i20 Sportz Diesel	2010	₹225,000	127,000	Diesel	Manual
Maruti Swift VXI BSIII	2007	₹130,000	120,000	Petrol	Manual

Observation: 5 seats is the median and most common seating value. Price, kilometres driven, engine and maximum power have wide ranges, so distribution shape and extreme observations matter when interpreting averages.

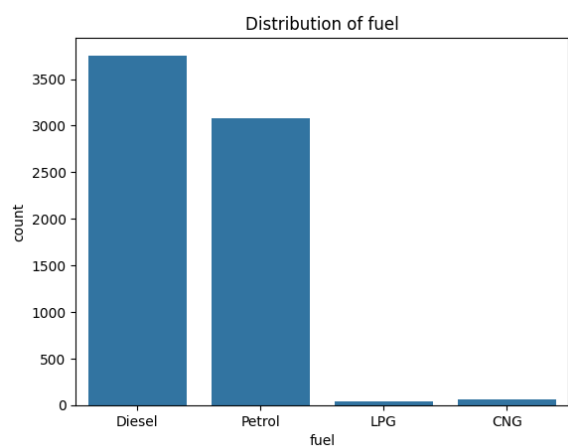
5. Visual EDA — Distributions and Categories

Chart 1. Distribution of Selling Price



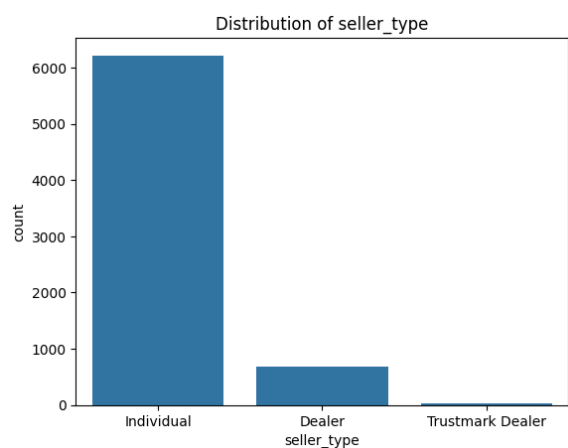
Observation: Selling price is strongly right-skewed: most observations are concentrated at lower prices, with a smaller group extending far into the high-price range.

Chart 2. Fuel Type Distribution



Observation: Diesel and Petrol account for the overwhelming majority of records, while LPG and CNG appear much less frequently.

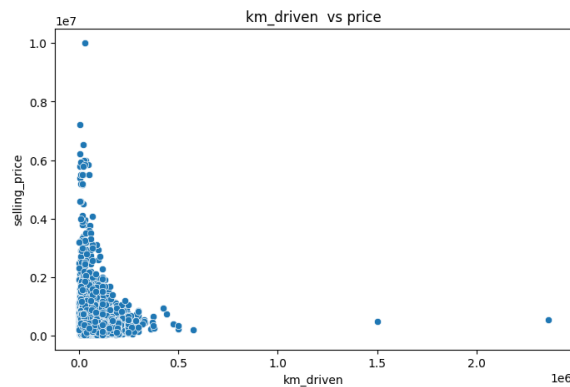
Chart 3. Seller Type Distribution



Observation: Individual sellers dominate the dataset, followed by dealers; Trustmark Dealers represent only a small fraction of records.

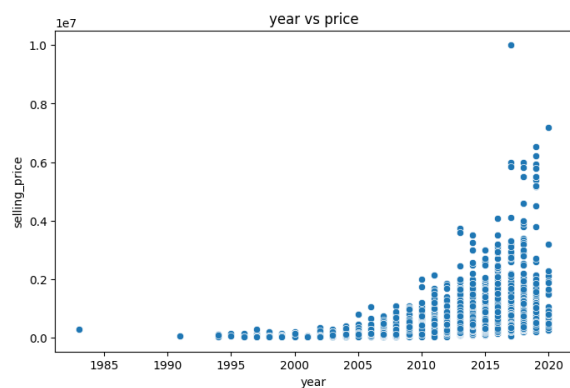
5. Visual EDA — Relationships and Correlations

Chart 4. Kilometres Driven vs Selling Price



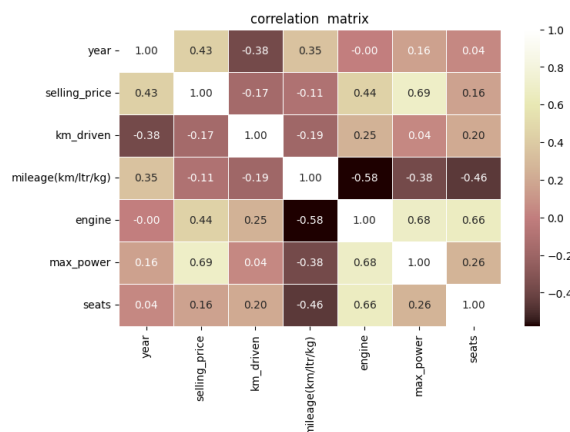
Observation: The plot shows a generally negative association: higher kilometres driven are often associated with lower selling prices, although substantial variation remains.

Chart 5. Vehicle Year vs Selling Price



Observation: Newer vehicles generally appear at higher selling prices than older vehicles. The relationship is not perfectly linear because other vehicle characteristics also vary.

Chart 6. Numerical Correlation Matrix



Observation: Selling price has a moderate positive correlation with max power (0.69) and engine size (0.44), and a moderate positive correlation with year (0.43). Km driven has a weak-to-moderate negative correlation with price (-0.17).

Chart source note: these figures are the visual outputs already recorded in the submitted notebook.

6. Key Findings

- **1. Selling price is highly skewed.** The mean price (about ₹517,271) is above the median (₹400,000), and the distribution contains a long upper tail.
- **2. Vehicle age is associated with price.** The year–price plot and correlation of about 0.43 indicate that newer vehicles tend to have higher selling prices in this dataset.
- **3. Maximum power has the strongest observed price correlation.** The correlation between max_power and selling_price is approximately 0.69.
- **4. Engine size also relates positively to price.** Engine and selling price have a correlation of approximately 0.44.
- **5. Higher usage is associated with lower price.** Km driven has a correlation of approximately -0.17 with selling price, consistent with the scatter plot.
- **6. A few categories dominate the sample.** Diesel, Individual seller type and Manual transmission are the most frequent categories in the notebook's descriptive output.

7. Limitations

- The notebook does not record a public dataset URL, so the report can only identify the source as cardekho.csv.
- Duplicate documentation is inconsistent: one output gives 0 duplicates, while a later comment says 1,202 duplicates were removed. The exact original duplicate count cannot be verified from the notebook alone.
- Sequential IQR filtering reduces the working dataset from 6,926 to 4,836 rows. Legitimate rare vehicles may also be removed, so findings describe the cleaned sample rather than the entire used-car market.
- The dataset does not contain every factor that can affect used-car prices, such as location, vehicle condition, accident history, service history, detailed trim and negotiation effects.
- Correlation represents association, not causation.

8. Conclusion

The EDA follows a practical workflow from inspection and data-quality checks through missing-value treatment, type correction, outlier handling, descriptive statistics and visual exploration. The analysis shows that used-car prices are highly variable and that year, engine size, maximum power and kilometres driven have observable relationships with price in this dataset. A key learning is that cleaning decisions can materially change the analysed sample, so row counts and cleaning rules should always be documented. Future work could compare price within similar vehicle groups, examine fuel and transmission effects more deeply, use robust statistics before and after outlier treatment, and develop a validated price-prediction model using an appropriately prepared dataset.

Submission alignment: The report includes dataset overview, data-quality checks, before/after cleaning information, descriptive statistics, six readable charts using multiple chart types, observations below each chart, evidence-based findings, limitations and conclusion. Full code is intentionally omitted.